Standard Operating Procedure for Resolving Critical Routing Protocol Flap ALARM_6154 on Arista Access Switches (Role: Access, Node ID: WFD-ACC-48, Site Location: Whitefield POP)

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6154

Description

This Standard Operating Procedure (SOP) is designed to guide the resolution of critical routing protocol flap failures, characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, on Arista access switches in the context of Node ID: WFD-ACC-48, situated at the Whitefield POP. The underlying architecture of these switches, leveraging the Arista Extensible Operating System (EOS), relies heavily on the Efficient Modular Architecture (EMA) and the Scalable Modularity for the next generation of data center networks. The EMA design ensures efficient packet processing and high-speed switching while the Scalable Modularity facilitates flexible configurations and seamless integration with existing network infrastructure. The routing protocol flaps in question can arise due to configuration inconsistencies, hardware issues, or software bugs. The flaps can further propagate and exacerbate the network instability, ultimately leading to critical alarms such as ALARM_6154. Therefore, it is essential to identify and rectify the root cause of these flaps to ensure stable and efficient network operation.

Pre-requisite Checks

- Verify that remote hands have been dispatched to the Whitefield POP to facilitate on-site intervention, if necessary.

- Check the status of redundant links and ensure that they are operational and configured correctly.

- Verify that the Arista switch is running the latest software version and that all EOS upgrades are current.

- Ensure that the network configuration is synchronized across all devices in the affected routing domain.

Potential Root Causes

Cause	Probability
Configuration inconsistency in BGP routing table	High
Hardware failure in the Arista access switch	Medium
Software bug in EOS causing ISIS LSP flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP dynamic peers: 1 BGP table version 1, BGP id 1.1.1.1 BGP router ID 1.1.1.1, local AS 65001, RIB entries 1 Peer groups: None, Description: This command is used to verify the BGP routing table and identify any inconsistencies that may be contributing to the flapping issue.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 1.1.1.1 128 FULL/ - 00:00:37 1.1.1.1 Eth1/1 1.1.1.2 128 FULL/ - 00:00:38 1.1.1.2 Eth1/2, Description: This command is used to verify the OSPF adjacency status and identify any issues that may be contributing to the flapping issue.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID LSP Seq Num LSP Checksum 1.1.1.1 0x80000001 0x12345678, Description: This command is used to verify the ISIS LSP flooding and identify any issues that may be contributing to the flapping issue.
```

Resolution Steps

Step 1: Step: Verify and Correct BGP Configuration, Command: show bgp summary, Description: Verify the BGP routing table and correct any configuration inconsistencies that may be contributing to the flapping issue.

Step 2: Step: Reset BGP Sessions, Command: clear bgp *, Description: Reset the BGP sessions to re-establish the connections and resolve the flapping issue.

Step 3: Step: Verify and Correct OSPF Configuration, Command: show ospf neighbor, Description: Verify the OSPF adjacency status and correct any configuration issues that may be contributing to the flapping issue.

Step 4: Step: Reset OSPF Sessions, Command: clear ospf *, Description: Reset the OSPF sessions to re-establish the connections and resolve the flapping issue.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP dynamic peers: 1 BGP table version 1, BGP id 1.1.1.1 BGP router ID 1.1.1.1, local AS 65001, RIB entries 1 Peer groups: None, Description: Verify that the BGP routing table is stable and consistent.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface 1.1.1.1 128 FULL/ - 00:00:37 1.1.1.1 Eth1/1 1.1.1.2 128 FULL/ - 00:00:38 1.1.1.2 Eth1/2, Description: Verify that the OSPF adjacency status is stable and consistent.

Command: show isis database, Expected Output: ISIS LSP Database: LSP ID LSP Seq Num LSP Checksum 1.1.1.1 0x80000001 0x12345678, Description: Verify that the ISIS LSP flooding is stable and consistent.
```

Roll-back Procedures

- Step: Restore Original Configuration, Command: copy startup-config running-config, Description: Restore the original configuration to roll back any changes made during the resolution process.

- Step: Reset Routing Protocols, Command: clear bgp *; clear ospf *, Description: Reset the routing protocols to their original state and re-establish the connections.